LUND UNIVERSITY

Written Examination
Group and Ring Theory
Friday May 28, 2021
Duration: 08.00–13.00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Solutions may of course be presented in Swedish! The sheets of the department should be used. Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

The solutions shall be uploaded in the form of a single *pdf-file in the Canvas system no later than 13:15. No collaboration or books are allowed. Questions concerning the formulation of the problems can be emailed to Arne Meurman* in the Canvas system. *I will check the mailbox at 09:00, 10:30, 12:00. The computer may only be used to read the problems, upload the solutions, and possibly email questions to Arne Meurman.*

1. Let A be the additive abelian group generated by x_1, x_2, x_3, x_4 subject to the relations

$$\begin{aligned} 7x_1 - 16x_2 + 23x_3 - 14x_4 &= 0, \\ -4x_1 + 10x_2 - 14x_3 + 8x_4 &= 0, \\ 2x_1 - 5x_2 + 7x_3 - 4x_4 &= 0. \end{aligned}$$

Express A as a direct sum of cyclic groups.

2. Let R be a ring with unity 1, and let I be a left ideal in R. Show that the following are equivalent:
(i) there exists a left ideal J such that $R = I \oplus J$ (as left R-modules),
(ii) there exists an idempotent $e \in I$ such that $I = Re$.

3. Let m be a positive integer and let A be an additive abelian group of order m. Consider A as a right $\mathbb{Z}$-module. Show that

$$A \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z} \simeq A.$$

4. Let M be a left R-module which has two decompositions

$$M = \bigoplus_{i=1}^{n} M_i = \bigoplus_{j=1}^{m} M_j^{'},$$

into sums of simple left modules $M_i, M_j^{'}$. Show that $n = m$ and there is a permutation π such that

$$M_i \simeq M_{\pi(i)}^{'},$$

for $1 \leq i \leq n$.

5. Let R be a ring, and let $I_1, \ldots, I_n$ be ideals in R such that each R/I_j is Noetherian and $I_1 \cap \cdots \cap I_n = (0)$. Prove that R is Noetherian.

6. Show that there is no simple group of order 400. (Burnsides Theorem may not be used without proof.)